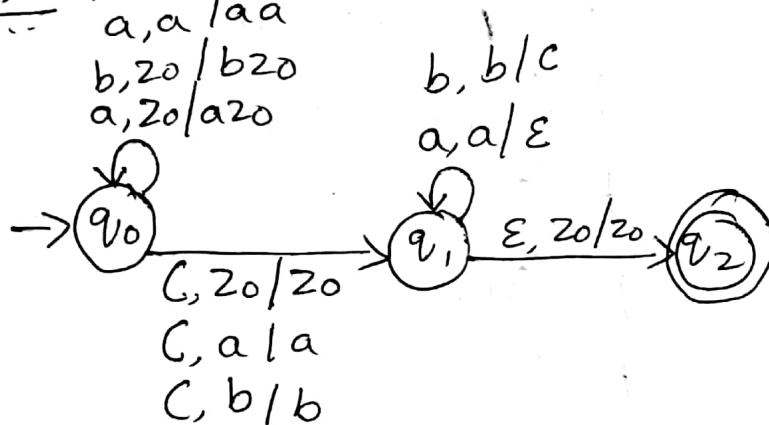


1) Design a PDA for the following language  
 $L = \{wcw^R : w \in \{a, b\}^*\}$

Also draw the transition diagram. write the computation (sequence of all configurations) for the input string 'abaCaba' and 'abCab'

Soln:  
 $b, a / ba$   
 $a, b / ab$   
 $b, b / bb$   
 $a, a / aa$   
 $b, z_0 / bz_0$   
 $a, z_0 / az_0$



$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

$$Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{a, b, C\}$$

$$\Gamma = \{a, b, z_0\}$$

$q_0 \in Q$  is the start state of machine  
 $z_0 \in \Gamma$  is the initial symbol on the stack  
 $F = \{q_2\}$  is the final state

$$\delta(q_0, a, z_0) = (q_0, az_0)$$

$$\delta(q_0, b, z_0) = (q_0, bz_0)$$

$$\delta(q_0, a, b) = (q_0, ab)$$

$$\delta(q_0, b, a) = (q_0, ba)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, b) = (q_0, bb)$$

$$\delta(q_0, C, z_0) = (q_1, z_0)$$

$$\delta(q_0, C, a) = (q_1, a)$$

$$\delta(q_0, C, b) = (q_1, b)$$

$$\delta(q_1, a, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, b) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$$

The sequence of moves made by the PDA for the string  $abacaba$  is shown below

Initial ID

$$\begin{aligned} (q_0, abacaba, z_0) &\vdash (q_0, baacaba, az_0) \\ &\vdash (q_0, aacaba, baz_0) \\ &\vdash (q_0, cacaba, abaz_0) \\ &\vdash (q_1, abacaba, abaz_0) \\ &\vdash (q_1, ba, baz_0) \\ &\vdash (q_1, a, az_0) \\ &\vdash (q_1, \epsilon, z_0) \\ &\vdash (q_2, \epsilon, z_0) \end{aligned}$$

(final Configuration)

Since  $q_2$  is the final state and input string is  $\epsilon$  in the final Configuration, the string  $abacaba$  is accepted by the PDA.

→ for the string abCab

Initial ID

$$\begin{aligned} (q_0, abCab, z_0) &\vdash (q_0, bCab, az_0) \\ &\vdash (q_0, Cab, baz_0) \\ &\vdash (q_1, \underset{\uparrow}{a}b, \underset{\uparrow}{b}az_0) \end{aligned}$$

(Final Configuration)

Since the transition  $\delta(q_1, a, b)$  is not defined, the string  $abCab$  is not a palindrome and the m/c halts and the string is rejected by the PDA.

Pushdown Automata

The formal definition of PDA is shown below:

Defn: A pushdown automata (PDA) is a seven tuple

$$M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, A)$$

where,

$Q$  - is set of finite states.

$\Sigma$  - Set of input alphabets

$\Gamma$  - Set of stack alphabets

$\delta$  - Transition from  $Q \times (\Sigma \cup \epsilon) \times \Gamma$  to finite sub set of  $Q \times \Gamma^*$ .

$\delta$  is called the transition function of  $M$

$q_0 \in Q$  is the start state of machine

$Z_0 \in \Gamma$  is the initial symbol on the stack

$A \subseteq Q$  is set of final states.

The actions performed by the PDA depends

On 1. The current state.

2. The next i/p symbol

3. The symbol on top of the stack.

The action performed by the machine consists of

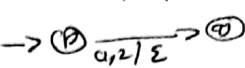
1. Changing the states from one state to another

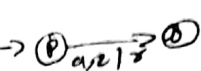
2. Replacing the symbol on the stack

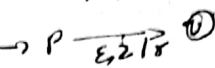
... of the further  
... but ...  
... ...

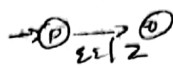
1) What does each of the following transitions represent? <sup>Corresponding Transition Diagram</sup>

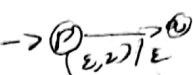
a.  $\delta(p, a, z) = (q, az)$   $\rightarrow$  

b.  $\delta(p, a, z) = (q, \epsilon)$   $\rightarrow$  

c.  $\delta(p, a, z) = (q, r)$   $\rightarrow$  

d.  $\delta(p, \epsilon, z) = (q, r)$   $\rightarrow$  

e.  $\delta(p, \epsilon, \epsilon) = (q, z)$   $\rightarrow$  

f.  $\delta(p, \epsilon, z) = (q, \epsilon)$   $\rightarrow$  

$\rightarrow$  The transition  $\delta(p, a, z) = (q, az)$  means that the PDA in current state  $p$  after scanning the i/p symbol  $a \in \Sigma$  and if  $z \in \Gamma$  is on top of the stack, then the PDA enters into new state  $q$  pushing  $a \in \Sigma$  on to the stack.

$\rightarrow$  The transition  $\delta(p, a, z) = (q, \epsilon)$  means that in state  $p$ , on scanning the i/p symbol  $a$ , and when top of this stack is  $z$ , the machine enters into state  $q$  and the topmost symbol  $z$  is deleted from the stack.

$\rightarrow$  The transition  $\delta(p, a, z) = (q, r)$  means that in state  $p$ , on scanning the input symbol  $a$  and when top of the stack is  $z$ , the machine enters into state  $q$  and topmost symbol  $z$  on the stack is replaced by  $r$ .

$\rightarrow$  The transition  $\delta(p, \epsilon, z) = (q, r)$  means that in state  $p$ , on scanning the Empty string and when top of the stack is  $z$ , the machine enters into state  $q$  and topmost symbol  $z$  on the stack is replaced by  $r$ .

→ The transition  $\delta(p, \epsilon, \epsilon) = (q, z)$  means that in state  $p$ , on scanning the Empty string and when top of the stack Empty, the machine enters into  $q$  pushing the symbol  $z$  on the stack. —

→ The transition  $\delta(p, \epsilon, z) = (q, \epsilon)$  means that in state  $p$  on scanning the Empty string and when top of the stack is  $z$ , the machine enters into  $q$  and topmost symbol  $z$  on the stack is deleted.



2) Obtain a PDA to accept the language  
 $L = \{a^n b^n \mid n \geq 1\}$  by a final state.

Soln

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, A)$$

where  $Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, z_0\}$

$\delta$  is

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, a, a) = (q_0, a a)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

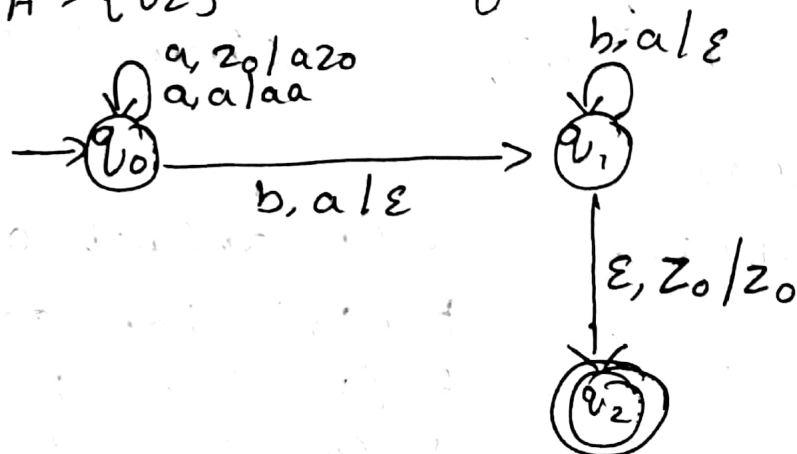
$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$$

$q_0 \in Q$  is the start state of m/c

$z_0 \in \Gamma$  is the initial symbol on the stack

$A = \{q_2\}$  is the final state



3) Obtain a PDA to accept the language  
 $L(M) = \{w \mid w \in (a+b)^* \text{ and } n_a(w) = n_b(w) \text{ by a final state}\}$   
 (i.e. number of a's in string  $w$  should be Equal to number of b's in  $w$ ). Also show the moves made by the PDA for the string abbaba

Soln:  
 $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, A)$

where

$Q = \{q_0, q_1\}$

$\Sigma = \{a, b\}$

$\Gamma = \{a, b, z_0\}$

$\delta$  is

$\delta(q_0, a, z_0) = (q_0, az_0)$

$\delta(q_0, b, z_0) = (q_0, bz_0)$

$\delta(q_0, a, a) = (q_0, aa)$

$\delta(q_0, b, b) = (q_0, bb)$

$\delta(q_0, a, b) = (q_0, \epsilon)$

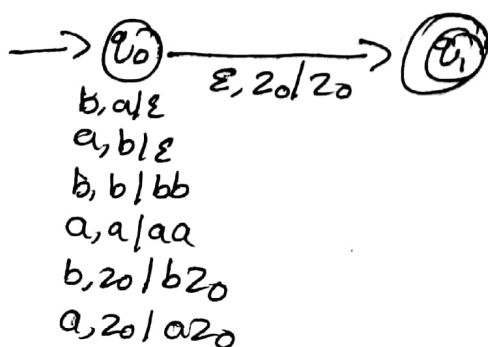
$\delta(q_0, b, a) = (q_0, \epsilon)$

$\delta(q_0, \epsilon, z_0) = (q_1, z_0)$

$q_0 \in Q$  is the start <sup>state</sup> symbol of m/c

$z_0 \in \Gamma$  is the initial symbol on the stack

$A = \{q_1\}$  is the final state



$(q_0, abbaba, z_0) \vdash (q_0, bbabaz_0, az_0)$

$\vdash (q_0, babaz_0, az_0)$

$\vdash (q_0, abaz_0, abz_0)$

$\vdash (q_0, baz_0, abz_0)$

$\vdash (q_0, a, abz_0)$

$\vdash (q_0, \epsilon, abz_0)$

$\vdash (q_1, \epsilon, abz_0)$

4) Obtain a PPA to accept a string of balanced parentheses. The parentheses to be considered are (, ), [, ].

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

where

$$Q = \{q_0, q_1\}$$

$$\Sigma = \{ (, ), [, ] \}$$

$$\Gamma = \{ (, [, z_0 \}$$

$$\delta_{10} \quad \delta(q_0, \epsilon, z_0) = (q_0, z_0)$$

$$\delta(q_0, (, z_0) = (q_1, (z_0)$$

$$\delta(q_0, [, z_0) = (q_1, [z_0)$$

$$\delta(q_1, (, ( ) = (q_1, (($$

$$\delta(q_1, (, [ ) = (q_1, ([$$

$$\delta(q_1, [, [ ) = (q_1, [[$$

$$\delta(q_1, [, ( ) = (q_1, [($$

$$\delta(q_1, ), ( ) = (q_1, \epsilon)$$

$$\delta(q_1, ], [ ) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_0, z_0)$$

$q_0 \in Q$  is the start state of machine.

$z_0 \in \Gamma$  is the initial symbol on the stack

$$F = \{q_0\}$$

5) Obtain a PPA to accept the language  $L = \{w \mid w \in (a, b)^*$  and  $n_a(a) = n_b(b)\}$

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, b, z_0) = (q_0, b z_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, b) = (q_0, bb)$$

$$\delta(q_0, a, b) = (q_0, \epsilon)$$

$$\delta(q_0, b, a) = (q_0, \epsilon)$$

$$\delta(q_0, \epsilon, a) = (q_1, a)$$

$$6) a^n b^{2n} \mid n \geq 1$$

$$\delta(q_0, a, z_0) = (q_0, a a z_0)$$

$$\delta(q_0, a, a) = (q_0, a a a)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, \epsilon)$$

$$7) L(m) = \{w w^R \mid w \in (a, b)^*\}$$

$$\delta(q_0, \epsilon, z_0) = (q_1, z_0)$$

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, b, z_0) = (q_0, b z_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, b) = (q_0, bb)$$

$$\delta(q_0, a, b) = (q_0, ab)$$

$$\delta(q_0, b, a) = (q_0, ba)$$

$$\delta(q_0, a, a) = (q_1, \epsilon)$$

$$\delta(q_0, b, b) = (q_1, \epsilon)$$

$$\delta(q_1, a, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, b) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$$

$$\delta(q_2, \epsilon, a) = (q_2, a)$$

~~aaaa~~  
abaaab



## Deterministic and NonDeterministic PDA

defn: Let  $M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, A)$  be a PDA. The PDA is deterministic if

1.  $\delta(q, a, z)$  has only one element.
2. If  $\delta(q, \epsilon, z)$  is not empty, then  $\delta(q, a, z)$  should be empty.

If one of the Condition fails, the PDA is non-deterministic.

1) Is the PDA to accept the language  $L(M) = \{wcw^R / w \in (a+b)^*\}$  is deterministic?

Solu:  
 $\delta(q_0, a, z_0) = (q_0, az_0)$   
 $\delta(q_0, b, z_0) = (q_0, bz_0)$   
 $\delta(q_0, a, a) = (q_0, aa)$   
 $\delta(q_0, b, a) = (q_0, ba)$   
 $\delta(q_0, a, b) = (q_0, ab)$   
 $\delta(q_0, b, b) = (q_0, bb)$   
 $\delta(q_0, c, z_0) = (q_1, z_0)$   
 $\delta(q_0, c, a) = (q_1, a)$   
 $\delta(q_0, c, b) = (q_1, b)$   
 $\delta(q_1, a, a) = (q_1, \epsilon)$   
 $\delta(q_1, b, b) = (q_1, \epsilon)$   
 $\delta(q_1, \epsilon, z_0) = (q_2, z_0)$   $(q_1, a, z_0) = \emptyset$

The PDA should satisfy the two conditions

1)  $\delta(q, a, z)$  has only one element: Note that in the transitions, for each  $q \in Q$ ,  $a \in \Sigma$  and  $z \in \Gamma$ , there is only one component defined and the first condition is satisfied.

2. The Second Condition states that if  $\delta(q, \epsilon, z)$  is not empty, then  $\delta(q, a, z)$  should be empty i.e. if there is an  $\epsilon$ -transition (in this case it is  $\delta(q_1, \epsilon, z_0)$ )) then there should not be any transitions from the state  $q_1$  when top of the stack is  $z_0$  which is true.

Since PDA satisfies both the conditions, the PDA is deterministic.

2) Is the PDA corresponding to the language  $L = \{a^n b^n / n \geq 1\}$  by a final state is deterministic?

$$\begin{aligned}\delta(q_0, a, z_0) &= (q_0, az_0) \\ \delta(q_0, a, a) &= (q_0, aa) \\ \delta(q_0, b, a) &= (q_1, \epsilon) \\ \delta(q_0, b, a) &= (q_1, \epsilon) \\ \delta(q_1, \epsilon, z_0) &= (q_2, \epsilon)\end{aligned}$$

deterministic

3) Is the PDA corresponding to accept the language  $L(M) = \{w / w \in (a+b)^*\}$  and  $n_a(w) = n_b(w)$  is deterministic?

$$\begin{aligned}\delta(q_0, a, z_0) &= (q_0, az_0) \\ \delta(q_0, a, a) &= (q_0, aa) \\ \delta(q_0, b, z_0) &= (q_0, bz_0) \\ \delta(q_0, b, a) &= (q_0, ab) \\ \delta(q_0, a, b) &= (q_0, \epsilon) \\ \delta(q_0, b, a) &= (q_0, \epsilon) \\ \delta(q_0, \epsilon, z_0) &= (q_1, z_0)\end{aligned}$$

1st condition satisfied  
2nd  $\rightarrow (q_0, \epsilon, z_0)$   
 $\rightarrow 1) q_0, a, z_0$  is defined so it.  
 $\rightarrow 2) q_0, b, z_0$   
non deterministic

4) Is the PDA to accept the language  $L = \{w / w \in (a, b)^* \text{ and } n_a(w) > n_b(w)\}$  is deterministic?

$$\begin{aligned}\delta(q_0, a, z_0) &= (q_0, az_0) \\ \delta(q_0, a, a) &= (q_0, aa) \\ \delta(q_0, b, z_0) &= (q_0, bz_0) \\ \delta(q_0, b, a) &= (q_0, \epsilon) \\ &= (q_0, bb)\end{aligned}$$

$$\begin{aligned}\delta(q_0, a, b) &= (q_0, \epsilon) \\ \delta(q_0, \epsilon, a) &= (q_1, a)\end{aligned}$$

$\delta(q_0, \epsilon, a) = (q_1, a)$   
the transition  $\delta(q_0, \epsilon, a)$  where  $\epsilon \in \Sigma$  should not be defined, But there two  
 $\delta(q_0, a, a) = (q_0, aa)$   $\delta(q_0, b, a) = (q_0, \epsilon)$   
so it is non deterministic

## CFG to PDA

Step 1: push the start symbol  $S$  of grammar  $G$  into the stack. The transition can be written as

$$\delta(\underline{p}, \underset{\uparrow}{\epsilon}, \underset{\uparrow}{\epsilon}) = (\underline{q}, \underline{S})$$

Step 2: For every production of the form  $A \rightarrow \alpha_1 \alpha_2 \dots \alpha_n$  introduce the following transition.

$$\delta(\underline{q}, \underline{\epsilon}, \underline{A}) = (\underline{q}, \alpha_1 \alpha_2 \alpha_3 \dots \alpha_n)$$

i.e., replace the symbol  $A$  on the stack with RHS of the production  $\alpha_1 \alpha_2 \dots \alpha_n$ .

So, if  $A \rightarrow \epsilon$  is a production, introduce the transition

$$\delta(\underline{q}, \underline{\epsilon}, \underline{A}) = (\underline{q}, \underline{\epsilon})$$

Step 3: for every  $a \in T$ , introduce the transition:

$$\delta(\underline{q}, \underline{a}, \underline{a}) = (\underline{q}, \underline{\epsilon})$$

1) Convert the following grammar into equivalent PDA

$$E \rightarrow E + T / T$$

$$T \rightarrow T * F / F$$

$$F \rightarrow (E) / id$$

Soln: ~~S~~  $S$  is shown below

$$\delta(p, \epsilon, \epsilon) = (q, E)$$

$$\delta(q, \epsilon, E) = (q, E + T)$$

$$\delta(q, \epsilon, E) = (q, T)$$

$$\delta(q, \epsilon, T) = (q, T * F)$$

$$\delta(q, \epsilon, T) = (q, F)$$

$$\delta(q, \epsilon, F) = (q, (E))$$

$$\delta(q, \epsilon, F) = (q, id)$$

$$\delta(q, +, +) = (q, \epsilon)$$

$$\delta(q, *, *) = (q, \epsilon)$$

$$\delta(q, (, () = (q, \epsilon)$$

$$\delta(q, id, id) = (q, \epsilon)$$

$$\delta(q, id, id) = (q, \epsilon)$$

$$M = (Q, \Sigma, \Gamma, \delta, q_0, F)$$

$$Q = \{p, q\}$$

$$\Sigma = \{id, +, *, (, )\}$$

$$q_0 = \{p\}$$

$$F = \{q\}$$

$$\Gamma = \{E, T, F, id, +, *, (, )\}$$

## CFE to PDA

→ For the grammar

$$S \rightarrow aAB B / aAA$$

$$A \rightarrow aBB / a$$

$$B \rightarrow bBB / A$$

$$C \rightarrow a$$

obtain the corresponding PDA

Soln:

$$M = (Q, \Sigma, \Gamma, \delta, q_0, F)$$

$\delta$  is shown below

$$\delta(p, \epsilon, \epsilon) = (q, S)$$

$$\delta(q, \epsilon, S) = (q, aAB B)$$

$$\delta(q, \epsilon, S) = (q, aAA)$$

$$\delta(q, \epsilon, A) = (q, aBB)$$

$$\delta(q, \epsilon, A) = (q, a)$$

$$\delta(q, \epsilon, B) = (q, bBB)$$

$$\delta(q, \epsilon, B) = (q, A)$$

$$\delta(q, \epsilon, C) = (q, a)$$

$$\delta(q, a, a) = (q, \epsilon)$$

$$\delta(q, b, b) = (q, \epsilon)$$

~~Q~~

$$Q = \{p, q\}$$

$$\Sigma = \{a, b\}$$

$$\Gamma = \{a, b, A, B, C, S\}$$

$q_0 = \{p\}$  is the start state

$F = \{q\}$  is the Final state



PDA to CFG -  $M = \{Q, \Sigma, \delta, q_0, \Gamma, F\}$   
 $G = (\underline{V}, \underline{\Gamma}, \underline{P}, \underline{S})$

1

- 1) The  $\Sigma$  symbols of PDA will be the terminals of CFG.
- 2) If the PDA moves from state  $q_i$  to state  $q_j$  on consuming the  $\Gamma$   $a \in \Sigma$ , when  $z$  is the top of the stack, then the non-terminals of CFG are the triplets of the form  $(q_i \underline{z} q_j)$ .
- 3) If  $q_0$  is the start state and  $q_f$  is the final state, then  $(q_0 \underline{z} q_f)$  is the start symbol of CFG.
- 4) The productions of CFG can be obtained from the transitions of PDA as shown below:

a) For Each transition of the form

$$\delta(q_i, a, z) = (q_j, AB)$$

introduce the productions of the form

$$q_i \underline{z} q_k \rightarrow a (q_j A q_l) (q_l B q_k)$$

where  $q_k$  and  $q_l$  will take all possible values from  $Q$ .

b) For Each transition of the form

$$\delta(q_i, a, z) = (q_j, \epsilon)$$

introduce the production

$$(q_i \underline{z} q_j) \rightarrow a$$

$$\delta(q_0, a, A) = (q_1, \epsilon)$$

$$q_0 A q_1 \rightarrow a$$

$$\delta(q_0, a, A) = (q_1, A^2)$$

$$q_0 A^2 q_1 = (a q_0)$$



## PDA to CFE

1) Obtain a CFE for the PDA shown below

$$\delta(q_0, a, z) = (q_0, Az)$$

$$\delta(q_0, a, A) = (q_0, A)$$

$$\delta(q_0, b, A) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z) = (q_2, \epsilon)$$

Soln.

$$\delta(q_0, a, z) = (q_0, Az)$$

$$\delta(q_0, a, A) = (q_3, \epsilon)$$

$$\delta(q_0, b, A) = (q_1, \epsilon)$$

$$\delta(q_3, \epsilon, z) = (q_0, Az)$$

$$\delta(q_1, \epsilon, z) = (q_2, \epsilon)$$

| $\delta(q_i, a, z) = (q_j, AB)$              | $q_i z q_k \rightarrow a(q_i A q_1)(q_1 B q_k)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\delta(q_0, a, z) = (q_0, Az)$              | $(q_0 z q_0) \rightarrow a(q_0 A q_0)(q_0 z q_0) / a(q_0 A q_1)(q_1 z q_0) /$<br>$a(q_0 A q_2)(q_2 z q_0) / a(q_0 A q_3)(q_3 z q_0)$<br>$(q_0 z q_1) \rightarrow a(q_0 A q_0)(q_0 z q_1) / a(q_0 A q_1)(q_1 z q_1) /$<br>$a(q_0 A q_2)(q_2 z q_1) / a(q_0 A q_3)(q_3 z q_1)$<br>$(q_0 z q_2) \rightarrow a(q_0 A q_0)(q_0 z q_2) / a(q_0 A q_1)(q_1 z q_2) /$<br>$a(q_0 A q_2)(q_2 z q_2) / a(q_0 A q_3)(q_3 z q_2)$<br>$(q_0 z q_3) \rightarrow a(q_0 A q_0)(q_0 z q_3) / a(q_0 A q_1)(q_1 z q_3) /$<br>$a(q_0 A q_2)(q_2 z q_3) / a(q_0 A q_3)(q_3 z q_3)$ |
| $\delta(q_3, \epsilon, z) = (q_0, Az)$       | $(q_3 z q_0) \rightarrow (q_0 A q_0)(q_0 z q_0) / (q_0 A q_1)(q_1 z q_0) /$<br>$(q_0 A q_2)(q_2 z q_0) / (q_0 A q_3)(q_3 z q_0)$<br>$(q_3 z q_1) \rightarrow (q_0 A q_0)(q_0 z q_1) / (q_0 A q_1)(q_1 z q_1) /$<br>$(q_0 A q_2)(q_2 z q_1) / (q_0 A q_3)(q_3 z q_1)$<br>$(q_3 z q_2) \rightarrow (q_0 A q_0)(q_0 z q_2) / (q_0 A q_1)(q_1 z q_2) /$<br>$(q_0 A q_2)(q_2 z q_2) / (q_0 A q_3)(q_3 z q_2)$<br>$(q_3 z q_3) \rightarrow (q_0 A q_0)(q_0 z q_3) / (q_0 A q_1)(q_1 z q_3) /$<br>$(q_0 A q_2)(q_2 z q_3) / (q_0 A q_3)(q_3 z q_3)$                 |
| $\delta(q_0, a, A) = (q_3, \epsilon)$        | $q_0 A q_3 \rightarrow a$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\delta(q_0, b, A) = (q_1, \epsilon)$        | $q_0 A q_1 \rightarrow b$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| $\delta(q_1, \epsilon, z) = (q_2, \epsilon)$ | $q_1 z q_2 \rightarrow \epsilon$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

→ obtain CFC for the PDA shown below

$$\delta(q_0, a, Z) = (q_0, AZ)$$

$$\delta(q_0, b, A) = (q_0, AA)$$

$$\delta(q_0, a, A) = (q_1, \epsilon)$$

Soln

$$\delta(q_0, a, Z) = (q_0, AZ) \left| \begin{array}{l} (q_0 Z q_0) \rightarrow a(q_0 A q_0)(q_0 Z q_0) / \\ a(q_0 A q_1)(q_1 Z q_0) \end{array} \right|$$

$$(q_0 Z q_1) \rightarrow a(q_0 A q_0)(q_0 Z q_1) / \\ a(q_0 A q_1)(q_1 Z q_1)$$

$$\delta(q_0, b, A) = (q_0, AA) \left| \begin{array}{l} (q_0 A q_0) \rightarrow b(q_0 A q_0)(q_0 A q_0) / b(q_0 A q_1)(q_1 A q_0) \\ (q_0 A q_1) \rightarrow b(q_0 A q_0)(q_0 A q_1) / b(q_0 A q_1)(q_1 A q_1) \end{array} \right|$$

$$\delta(q_0, a, A) = (q_1, \epsilon) \left| q_0 A q_1 \rightarrow a \right|$$


---

## CFG to CNF

→ Consider the grammar

$$S \rightarrow 0A \mid 1B$$

$$A \rightarrow 0AA \mid 1S \mid 1$$

$$B \rightarrow 1BB \mid 0S \mid 0$$

obtain the grammar in CNF

Soln CNF for the given grammar is shown below

$$S \rightarrow A_0 A \mid A_1 B$$

$$A \rightarrow A_1 C \mid A_1 S \mid 1$$

$$B \rightarrow A_1 D \mid A_0 S \mid 0$$

$$A_0 \rightarrow 0$$

$$A_1 \rightarrow 1$$

$$C \rightarrow AA$$

$$D \rightarrow BB$$

→ Consider the grammar

$$S \rightarrow aBa \mid abba$$

$$A \rightarrow ab \mid AA$$

$B \rightarrow aB \mid a$  obtain the grammar in CNF

Soln: i) There is no  $\epsilon$  production

ii) There is no unit production

iii) eliminate useless productions & symbols, the resulting grammar is shown below (A is not reachable from start symbol so eliminate variable A & its productions)

$$S \rightarrow aBa \mid abba$$

$$B \rightarrow aB \mid a$$

iv) CNF is shown below

$$S \rightarrow CE \mid DE$$

$$B \rightarrow CB \mid a$$

$$C \rightarrow a$$

$$E \rightarrow BC$$

$$F \rightarrow b$$

$$D \rightarrow CF$$

$$C \rightarrow FC$$

## Greibach normal form (GNF):

Let  $G = (V, T, P, S)$  be a CFA is said to be in GNF if all the productions are of the form.

$$A \rightarrow a\alpha$$

where  $a \in T$  and  $\alpha \in V^*$  i.e. First symbol on the RHS of the production must be a terminal and it can be followed by zero or more variables.

1) Convert the following grammar

$$S \rightarrow AB1|0$$

$$A \rightarrow 00A|B$$

$$B \rightarrow 1A1 \text{ into GNF}$$

Soln:- 1) Eliminate ~~unit~~  $\epsilon$ , and unit production & useless production

2) Convert into ENF

3) finally Convert into GNF

$$* S \rightarrow AB1|0$$

$$A \rightarrow 00A|1A1 \quad \text{eliminated unit production}$$

$$B \rightarrow 1A1$$

$$* S \rightarrow ABA_1|0$$

$$A \rightarrow A_0 A_0 A | A_1 A A_1 \Rightarrow$$

$$B \rightarrow A_1 A A_1$$

$$A_0 \rightarrow 0$$

$$A_1 \rightarrow 1$$

$$S \rightarrow AC_1|0$$

$$A \rightarrow A_0 C_2 | A_1 C_3$$

$$B \rightarrow A_1 C_3$$

$$A_0 \rightarrow 0$$

$$A_1 \rightarrow 1$$

$$C_1 \rightarrow BA_1$$

$$C_2 = A_0 A$$

$$C_3 = AA_1$$

is in CNF for

$$\text{GNF} = \left\{ \begin{array}{l} S \rightarrow 0C_2C_1 | 1C_3C_1 | 0 \\ A \rightarrow 0C_2 | 1C_3 \\ B \rightarrow 1C_3 \\ A_0 \rightarrow 0 \\ A_1 \rightarrow 1 \\ C_1 \rightarrow 1C_3A_1 \\ C_2 \rightarrow 0A \\ C_3 = 0C_2A_1 | 1C_3A_1 \end{array} \right.$$

# Pumping Lemma for Context Free Languages:

Let  $L$  be the Context free language and is infinite.  
 Let  $Z$  is sufficiently long string and  $Z \in L$  so that  
 $|Z| > n$  where  $n$  is some positive integer. If the  
 String  $Z$  can be decomposed into combinations of  
 strings.

$$Z = uvwxy$$

Such that  $|vwx| \leq n$ ,  $|vx| > 1$ , then  $uv^iwx^iy \in L$  for  $i=0,1,2,\dots$

1) Show that  $L = \{a^n b^n c^n \mid n \geq 0\}$  is not Context free.

Step 1: Let  $L$  is Context free and is infinite. Let  $Z = a^n b^n c^n$

Step 2: Note that  $|Z| > n$  and so we can split  $Z$  into  
 $uvwxy$  such that

$$|vwx| \leq n \text{ and } |vx| > 1$$

and so  $uv^iwx^iy \in L$  for  $i=0,1,2,\dots$

Case 1: The string  $vwx$  is within  $a^n$

Let  $v = a^j$ ,  $x = a^k$  where  $|vx| = j+k \geq 1$  and  $|vwx| = n$   
 which can be shown as

$$Z = \underbrace{a \dots a}_n \underbrace{b \dots b}_n \underbrace{c \dots c}_n$$

$uvwx$   $y$

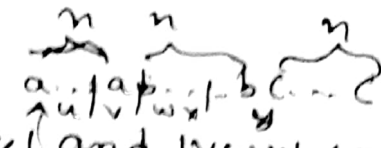
Now according to pumping lemma,  $uv^2wx^2y \in L$   
 for  $i=2$  and the language generated is as  
 shown below.

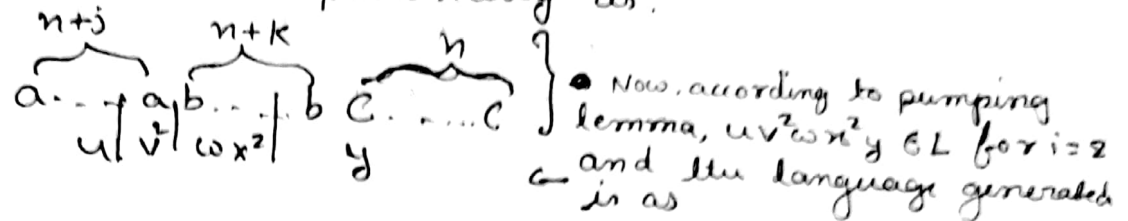
$$uv^2wx^2y = \underbrace{a \dots a}_{n+j+k} \underbrace{b \dots b}_n \underbrace{a \dots a}_n$$

$uv^2wx^2$   $y$



Note that  $uv^2wx^2y = a^{n+j+k} b^n c^n = \notin L$  when  $j+k \geq 1$ .  
But according to pumping lemma  $uv^2wx^2y \in L$ , which is the Contradiction.

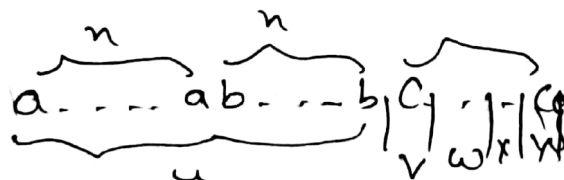
Case 2: The string  $vwx$  is  $a^n b^n$    
Let  $x = a^j$ ,  $x = b^k$  where  $|vx| = j+k \geq 1$  and  $|vwx| \leq n$   
which can be shown pictorially as:



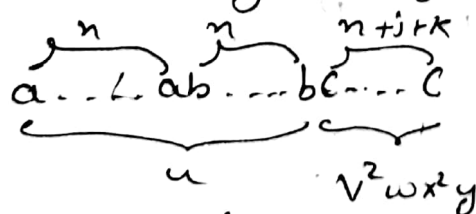
Note that  $uv^2wx^2y = a^{n+j} b^{n+k} c^n = \notin L$  when  $j+k \geq 1$ .  
But according to pumping lemma  $uv^2wx^2y \in L$  which is the Contradiction.

Case 3: The string  $vwx$  is either  $c^n$

Let  $V = c^j$ ,  $X = c^k$  where  $|vx| = j+k \geq 1$  and  $|vwx| \leq n$  which can be shown pictorially as



Now according to pumping lemma  $uv^2wx^2y \in L$  for  $i=2$  and the language generated is as shown below:



$uv^2wx^2y = a^n b^n c^{n+j+k} \notin L$  when  $j+k \geq 1$ . But According to pumping lemma  $uv^2wx^2y \in L$ , which is the Contradiction.

But, according to pumping lemma,  $n$  number of  $a$ 's should be followed by  $n$  number of  $b$ 's which in turn should be followed by  $n$  number of  $c$ 's. In all 3 cases we get Contradiction to the assumption that the language is Context free. So the language  $L = \{a^n b^n c^n / n \geq 0\}$  is not Context-free.

## Closure properties of Context free languages

CFL's are closed under Union, Concatenation & Star

Theorem: If  $L_1$  and  $L_2$  are CFLs, then  $L_1 \cup L_2$ ,  $L_1 L_2$  and  $L_1^*$  also denote the CFLs and so the Context free languages are closed under union, Concatenation, Star-closure.

Proof:- Let  $L_1$  and  $L_2$  are two CFLs generated by the CFGs.

$$G_1 = (V_1, T_1, P_1, S_1)$$

and

$$G_2 = (V_2, T_2, P_2, S_2)$$

respectively and assume that  $V_1$  and  $V_2$  are disjoint.

Case 1: Union of two CFLs is CFL

Now, let us consider the language  $L_3$  generated by the grammar.

$$G_3 = (V_1 \cup V_2 \cup S_3, T_1 \cup T_2, P_3, S_3)$$

where

- $S_3$  is a the start symbol for the grammar  $G_3$  and  $S_3 \in (V_1 \cup V_2)$
- $P_3 = P_1 \cup P_2 \cup \{S_3 \rightarrow S_1, S_2\}$

it is clear from this that the grammar  $G_3$  is Context free and the language generated by this grammar is Context free. it is Easy to prove that  $L_3 = L_1 \cup L_2$

If we assume  $w \in L_1$ , then the possible derivation from  $S_3$  is

$$S_3 \Rightarrow S_1 \xRightarrow{*} w$$

on similar lines if we assume  $w \in L_2$ , then the possible derivation from  $S_3$  is

$$S_3 \Rightarrow S_2 \xRightarrow{*} w$$

So if  $w \in L_3$ , one of the derivations

$$S_3 \Rightarrow S_1$$

$$\text{or } S_3 \Rightarrow S_2$$

is possible. In the first case, all the variables in  $V_1$  and all the terminals in  $T_1$  may be used to get the derivation

$$S_1 \xRightarrow{*} w$$

which uses only the productions in  $P_1$ . Similarly all the variables in  $V_2$  and all the terminals in  $T_2$  may be used to get the derivation

$$S_2 \xRightarrow{*} w$$

which uses only the productions in  $P_2$  and it follows that

$$L_3 = L_1 \cup L_2$$

Thus it is proved that CFL are closed under Union.

Concatenation of two CFLs is CFL

Now let us consider the language  $L_4$  generated by the grammar

$$G_4 = (V_1 \cup V_2 \cup S_4, T_1 \cup T_2, P_4, S_4)$$

where

- $S_4$  is a start symbol for the grammar  $G_4$  and  $S_4 \notin (V_1 \cup V_2)$

$$- P_4 = P_1 \cup P_2 \cup \{S_4 \rightarrow S_1 S_2\}$$

It is clear from this that the grammar  $G_4$  is Context free and the language generated by this grammar is Context free and so.

$$L_3 = L_1 L_2$$

Thus, it is proved that CFL are closed under Concatenation.

CFLs are closed under star-closure

Now, let us consider the language  $L_5$  generated by the grammar

$$G_5 = (V_1 \cup S_5, T_1, P_5, S_5)$$

Where

- $S_5$  is a start symbol for the grammar  $G_5$
- $P_5 = P_1 \cup \{S_5 \rightarrow S_1 S_5 | \epsilon\}$

it is clear from this that the grammar  $G_5$  is Context free and the language generated by this grammar is Context free and so

$$L_5 = L_1^*$$

Thus, it is proved that CFL are closed under Star-closure.

CFLs are not closed under intersection

Theorem: The CFLs are not closed under intersection. If  $L_1$  and  $L_2$  are Context free languages, it is not always true that  $L_1 \cap L_2$  is Context free language.

Proof: Let us prove this theorem by Counter Examples.

Consider the two languages.

$$L_1 = \{a^n b^n c^m \mid n \geq 0, m \geq 0\}$$

and 
$$L_2 = \{a^n b^m c^m \mid n \geq 0, m \geq 0\}$$



The two languages are Context free, as we can easily obtain the corresponding Context free grammars.

$$\begin{aligned} S &\rightarrow S_1 S_2 \\ S_1 &\rightarrow a S_1, b | \epsilon \\ S_2 &\rightarrow c S_2 | \epsilon \end{aligned}$$

and

$$\begin{aligned} S &\rightarrow a S_1, \\ S_1 &\rightarrow b S_1, c | \epsilon \end{aligned}$$

Now let us take

$$L_1 \cap L_2 = \{a^n b^n c^n / n \geq 0\}$$

This language is not Context free. So it is not closed under intersection.

### CFLs are not closed under Complementation

Theorem: The CFLs are not closed under Complementation. If  $L$  is Context free language, it is not true that Complement of  $L$  is Context free language.

proof: Let us prove this theorem by Contradiction. Suppose that Context free languages are closed under Complementation. So, if  $L_1$  and  $L_2$  are Context free languages, then  $\bar{L}_1$  and  $\bar{L}_2$  are also Context free. We have already proved that CFLs are closed under Union.

$$\text{So, } \bar{L}_1 \cup \bar{L}_2$$

must be Context free. Since we have assumed that the CFLs are closed under Complementation,

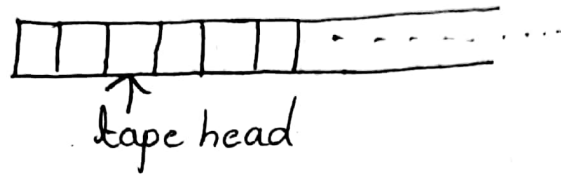
must be Context free. But according to De Morgan's law  $\overline{\bar{L}_1 \cup \bar{L}_2} = L_1 \cap L_2$



So,  $L_1 \cap L_2$  must be Context free which is a Contradiction. Since the Context free languages are not closed under intersection, our assumption that the CFLs are closed under Complementmentation is not true. So, the family of Context free languages are closed under Complementmentation.

# Turing machine

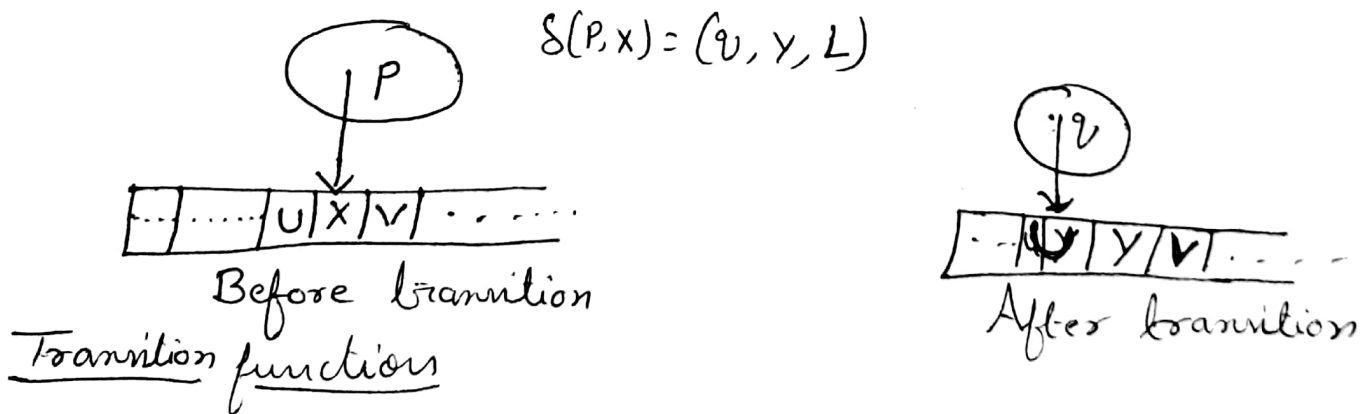
- Introduced by Alan Turing in 1936
- Finite Control with a tape
- Tape is unbounded and any cell of the tape can be accessed by using a tape head.



- A Turing m/c has a designated accept and reject state
- Transition of a Turing machine

$$\delta(p, x) = (q, y, L)$$

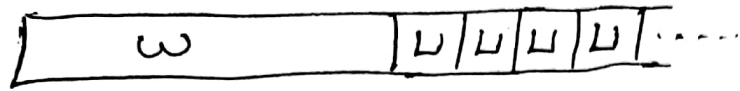
↑      ↑      ↑      ↑  
 State   tape symbol   state   tape symbol (replace x by y)   movement of the tape head (L)



$$\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$$

- Input is written on the tape itself
- Input alphabet  $\Sigma$  is a subset of the tape alphabet,  $\Gamma$   
 $\Sigma \subseteq \Gamma$

- There is a blank symbol,  $\sqcup \in \Gamma - \Sigma$
- Initially the tape contains the input  $w$  and the remaining cells are filled up with symbol  $\sqcup$



- If the input head tries to move to the left of the leftmost cell of the tape, then it stays at its current position.

Definition of Turing m/c: A Turing m/c (TM) is the tuple  $M = (\mathcal{Q}, \Sigma, \Gamma, \delta, q_0, q_A, q_R)$

where,

$\mathcal{Q}$  is the set of states

$\Sigma$  is the input alphabet

$\Gamma$  is the tape alphabet

$$\Sigma \subseteq \Gamma \text{ \& \exists, } \sqcup \in \Gamma \setminus \Sigma \quad (\setminus (-) \text{ minus sign})$$

$$\delta: \mathcal{Q} \times \Gamma \rightarrow \mathcal{Q} \times \Gamma \times \{L, R\}$$

$q_0$  is the start state

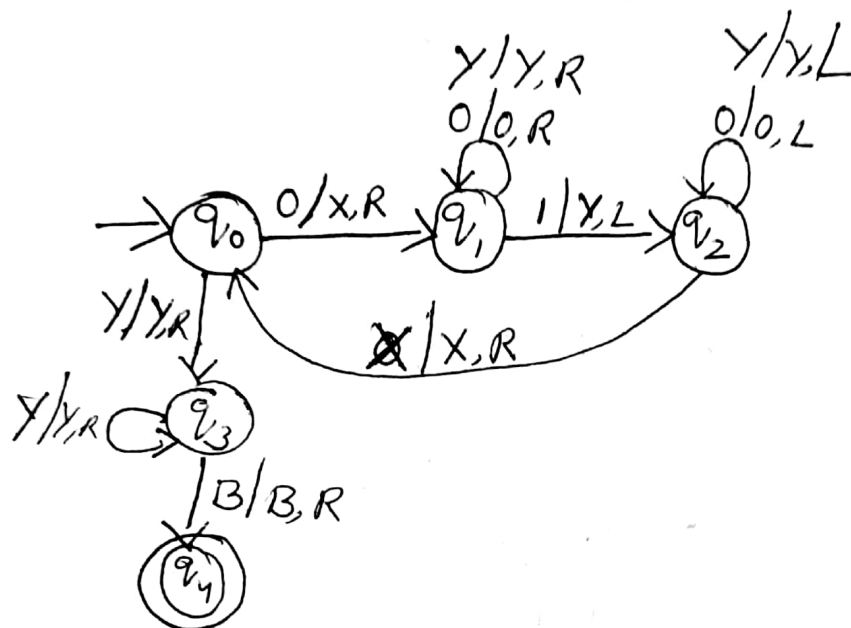
$q_A$  is the accept state

$q_R$  is the reject state.

→ obtain a Turing machine to accept the language  
 $L = \{0^n 1^n \mid n \geq 1\}$

Soln:

$$M = \{Q, \Sigma, \Gamma, q_0, \delta, B, F\}$$



$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

$$\Gamma = \{0, 1, x, y, B\}$$

$q_0 \in Q$  is the start state of machine

$B \in \Gamma$  is the blank symbol

$F = \{q_4\}$  is the final state

$\delta$  is shown below

$$\delta(q_0, 0) = (q_1, x, R)$$

$$\delta(q_1, 0) = (q_1, 0, R)$$

$$\delta(q_1, y) = (q_1, y, R)$$

$$\delta(q_1, 1) = (q_2, y, L)$$

$$\delta(q_2, 0) = (q_2, 0, L)$$

$$\delta(q_2, y) = (q_2, y, L)$$

$$\delta(q_2, x) = (q_0, x, R)$$

$$\delta(q_0, y) = (q_3, y, R)$$

$$\delta(q_3, y) = (q_3, y, R)$$

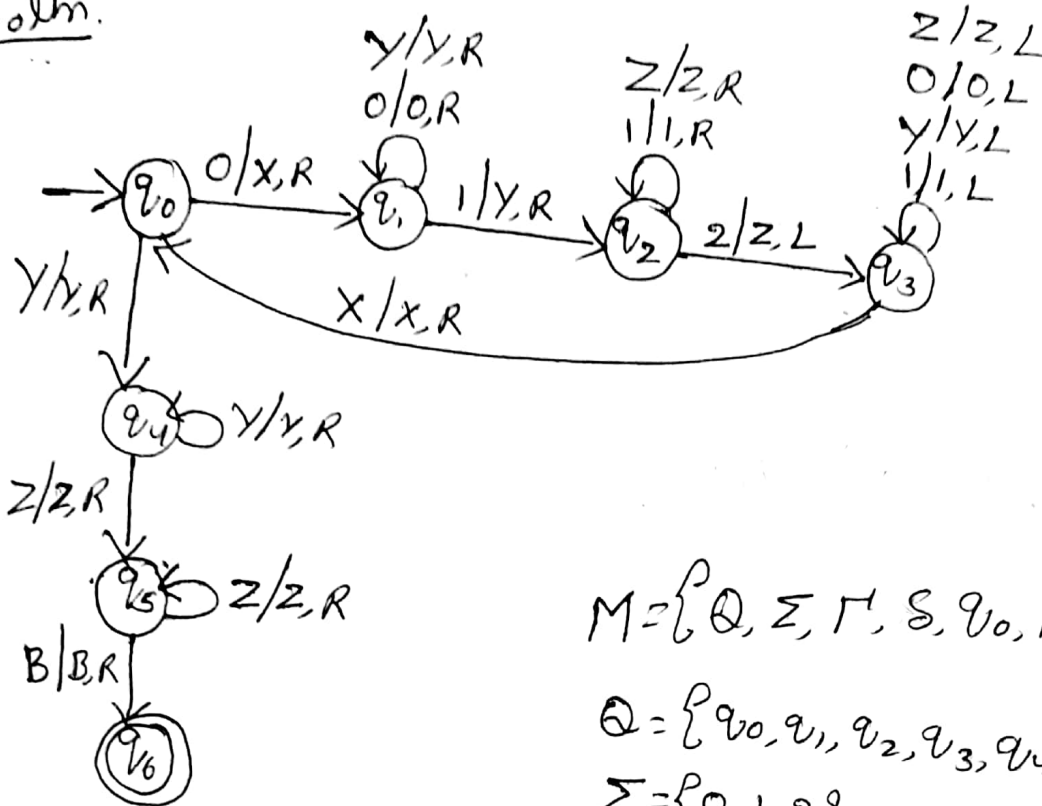
$$\delta(q_3, B) = (q_4, B, R)$$

| States | Tape Symbols ( $\Gamma$ ) |               |               |               |               |
|--------|---------------------------|---------------|---------------|---------------|---------------|
|        | 0                         | 1             | x             | y             | B             |
| $q_0$  | $(q_1, x, R)$             |               |               | $(q_3, y, R)$ |               |
| $q_1$  | $(q_1, 0, R)$             | $(q_2, y, L)$ |               | $(q_1, y, R)$ |               |
| $q_2$  | $(q_2, 0, L)$             |               | $(q_0, x, R)$ | $(q_2, y, L)$ |               |
| $q_3$  |                           |               |               | $(q_3, y, R)$ | $(q_4, B, R)$ |
| $q_4$  |                           |               |               |               |               |

Obtain a Turing machine to accept the language

$$L(M) = \{0^n 1^n 2^n \mid n \geq 1\}$$

Soln:



$$M = \{Q, \Sigma, \Gamma, \delta, q_0, F, B\}$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6\}$$

$$\Sigma = \{0, 1, 2\}$$

$$\Gamma = \{0, 1, 2, x, y, z, B\}$$

$q_0 \in Q$  is the start state of the machine  
 $F \in \{q_6\}$  is the Final state  
 $B$  is blank character

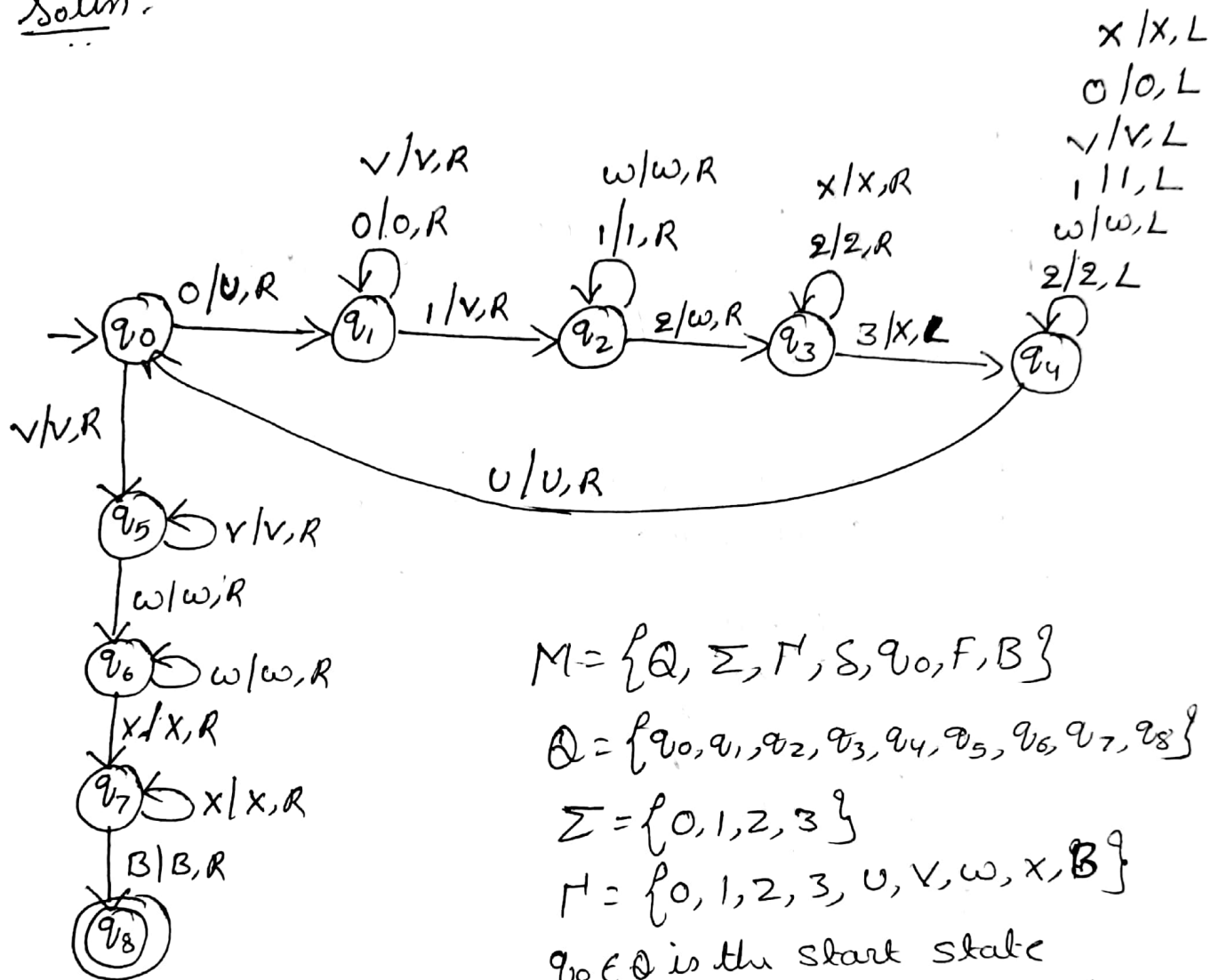
is shown below using  
Transition table

| States | $\Gamma$      |               |               |               |               |               |               |
|--------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|        | 0             | 1             | 2             | <del>z</del>  | y             | x             | B             |
| $q_0$  | $(q_1, x, R)$ |               |               |               | $(q_4, y, R)$ |               |               |
| $q_1$  | $(q_1, 0, R)$ | $(q_2, y, R)$ |               |               | $(q_1, y, R)$ |               |               |
| $q_2$  |               | $(q_2, 1, R)$ | $(q_3, z, L)$ | $(q_2, z, R)$ |               |               |               |
| $q_3$  | $(q_3, 0, L)$ | $(q_3, 1, L)$ |               | $(q_3, 2, L)$ | $(q_3, y, L)$ | $(q_0, x, R)$ |               |
| $q_4$  |               |               |               | $(q_5, z, R)$ | $(q_4, y, R)$ |               |               |
| $q_5$  |               |               |               | $(q_5, z, R)$ |               |               | $(q_6, B, R)$ |
| $q_6$  |               |               |               |               |               |               |               |



→ obtain a Turing machine to accept the language  
 $L = \{0^n 1^n 2^n 3^n \mid n \geq 1\}$

Soln:



$$M = \{Q, \Sigma, \Gamma, \delta, q_0, F, B\}$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_8\}$$

$$\Sigma = \{0, 1, 2, 3\}$$

$$\Gamma = \{0, 1, 2, 3, U, V, W, X, B\}$$

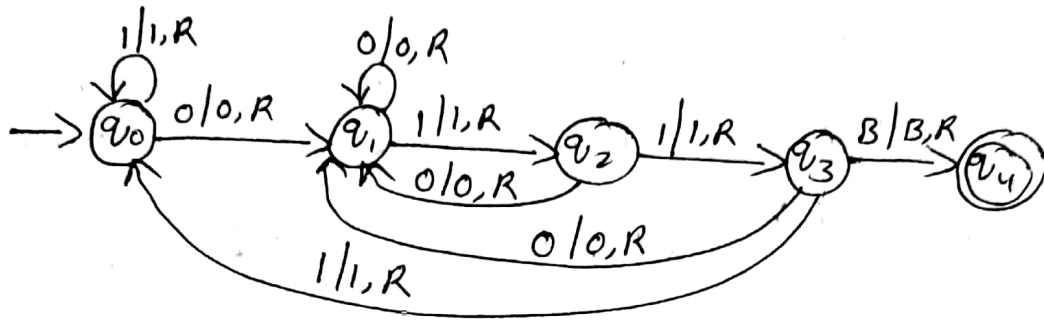
$q_0 \in Q$  is the start state

$F = \{q_8\}$  is the Final state

$B$  is blank character.

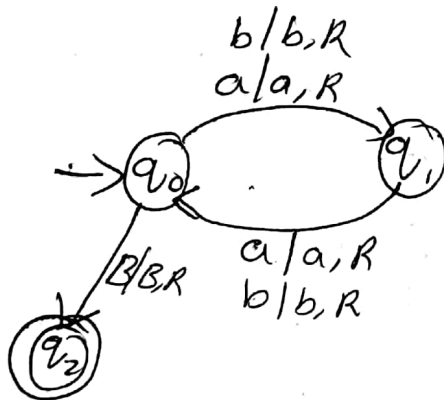
→ obtain a TM to accept the language  
 $L = \{w \mid w \in (0+1)^* \text{ Ending with } 011\}$

Soln:



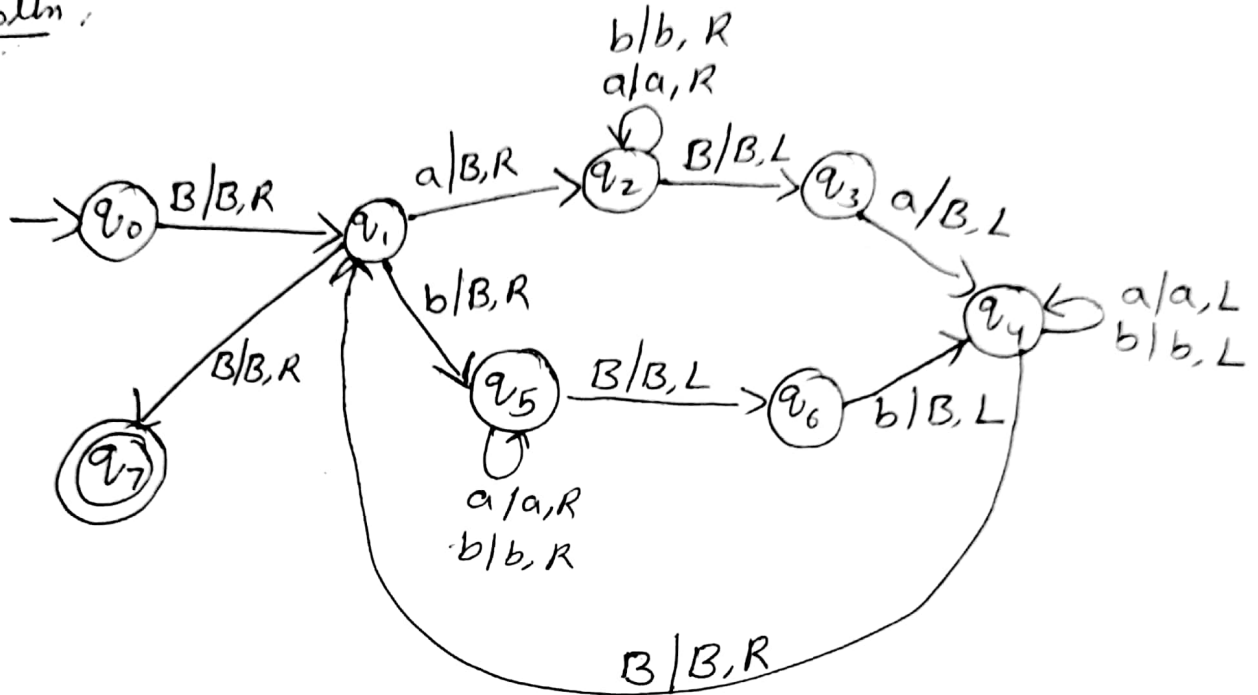
→ obtain a TM to accept the language  
 $L = \{w \mid w \text{ is even and } \Sigma = \{a, b\}\}$

Soln:



→ obtain a TM to accept the language  
 $L = \{ww^R \mid w \in (a+b)^*\}$

Soln:



→ obtain a Turing machine to accept a palindrome consisting of a's and b's of any length

Soln:

